

Article

Direct Adaptive Multi-Resampling Turbo Equalizer for Underwater Acoustic Single-Carrier Communication

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Abstract: A wideband Doppler Effect is a significant challenge for underwater acoustic communications (UAC). This paper proposes a new two-stage structure of direct adaptive multi-resampling turbo equalizer (DAM-TEQ) for solving the problem of large timescale errors in time-varying channels, which uses an innovative adaptive time-domain resampling method for Doppler estimation and compensation. In this equalizer, the received signal is first fed into the first-stage structure, in which an adaptive resampling is performed using equalization coefficient detection to achieve a Doppler rough estimation. After the processing is completed, it is fed into the second-stage structure for joint equalization and decoding, effectively reducing the error of information transmission. Compared with the conventional turbo equalizer (TEQ) based on timescale estimation, the proposed equalizer can avoid the problem of the Doppler Effect not being accurately estimated in time-varying channels, with only a slight increase in complexity. Simulations and lake trails show that the equalizer can effectively perform a Doppler estimation and compensation in time-varying channels, and has a better bit error rate (BER) performance than the traditional timescale-based TEQ.

Keywords: underwater acoustic communication; direct adaptive multi-resample; turbo equalizer



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1. Introduction

Underwater acoustic communication technology is widely used in civilian activities and military activities. Therefore, reliable UAC technology is of great importance to national maritime rights and interests [1]. However, the UAC channel is complex due to multipath, time variation, a limited bandwidth, and high environmental noise. At the same time, because of the slow transmission speed of underwater sound waves, the transceiver's relative motion and water flow during signal transmission will lead to a serious Doppler Effect [2–5]. In order to ensure the reliability of the communication system, the receiver must accurately estimate and compensate for the Doppler shift [6]. Effective methods to solve the Doppler Effect can be roughly divided into several categories: time-domain resampling [7–10], sparse channel estimation [11–16], and channel equalization [17–31].

Time-domain resampling algorithms such as match filter and time-domain cross-correlation use auxiliary or correlated signals to estimate the Doppler factor. Most of these methods are based on resampling to deal with timescale changes in UAC, combined with ambiguity functions [7], synchronous signals [8], and two-step methods [9,10]. One of the most classic algorithms takes advantage of the fact that the linear frequency modulation (LFM) signal is not susceptible to Doppler shift and has good autocorrelation. It estimates the Doppler factor (scale) through the correlation results of two LFM signals, which is less complex and easy to implement [8]. However, due to insufficient bandwidth and the limitation of the sampling rate in the communication process, the temporal resolution of the signal is poor, resulting in the ambiguity of the relevant peaks in the calculated output and the inability to identify them accurately. Moreover, due to the influence of a complex time-varying channel such as the multipath effect and attenuation, there is a possibility

of multiple correlation peaks. These conditions lead to the failure of traditional timescale-based estimation methods. Figure 1 shows the cross-correlation between the received signal and the LFM signal, which can be used for synchronization and time estimation by judging where the maximum peak occurs before and after the signal. As shown in the highlighted parts of the figure, the pre-correlated signal has an obvious peak. However, due to the influence of the time-varying channel, the peak of the post-correlated signal is not obvious, and even multiple correlated peaks appear, resulting in the inability to accurately estimate the time, which seriously affects the performance of UAC.

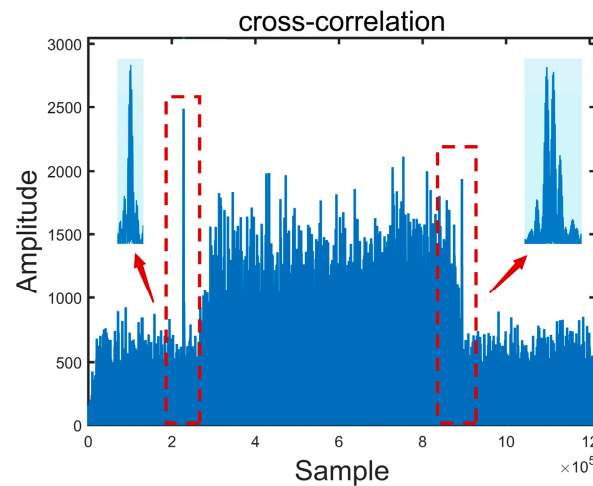


Figure 1. The cross-correlation results of the signal.

In addition, sparse channel estimation algorithms model the Doppler estimation as a representation and reconstruction of sparse signals [11]. Using the sparse channel estimation technique based on delay-Doppler-spread function representation, the Doppler factor can be estimated using the correlation principle and sparse reconstruction [12,13]. The two-stage sparse channel estimation technique estimates the delay and Doppler scale sequentially, greatly reducing the computational complexity [14]. A novel orthogonal matching pursuit approach is proposed to quickly explore time-varying sparsity in a two-step sequential manner [15]. However, accurate channel parameter estimation plays an important role in these methods. At the same time, there are relatively high requirements for the sparsity of underwater acoustic channels, which have certain limitations. In addition, a new Doppler compensation method is proposed in [16] based on time reversal focusing. Still, it is unsuitable for the actual system because of the high requirements for the communication equipment and the cumbersome process.

Combining the above methods with channel equalization technology can effectively alleviate the impact of the Doppler Effect and obtain better communication results. Various equalizers have been proposed, such as a coherent receiver structure based on a Decision Feedback Equalizer (DFE) [17] and a turbo equalizer [18]. Furthermore, some improved methods such as minimum mean square error turbo equalization (MMSE-TEQ) [19,20] and the direct adaptive turbo equalizer (DA-TEQ) [21,22] have been proposed sequentially. They both have a low complexity. Among them, the DA-TEQ is widely used in communication systems because of its low complexity and simple implementation. These methods are suitable for the correction and compensation of small Doppler shifts. However, due to the constraint of equalizer convergence, the Doppler Effect can be compensated for in a small range. On this basis, a commonly used conventional TEQ based on timescale estimation first uses resampling technology to compensate for Doppler, and then sends the compensated signal to the TEQ for processing to correct the residual Doppler. This approach is effective in mitigating the Doppler Effect, but at the same time, it is still inevitably limited by the channel and the associated signal.

As described, time-domain resampling uses signal expansion or compression on the timescale for Doppler estimation. It has a low complexity and is widely used. However, good channels and accurate timescales are required. Sparse channel estimation requires a high level of complexity, while maintaining a high accuracy. In addition, it also requires the sparsity of the channel. Channel equalization can effectively address small-scale Doppler Effects, but it fails with larger scales. Therefore, combined with channel equalization technology [23–31], this paper proposes a DAM-TEQ, and uses the adaptive time-domain resampling method for Doppler estimation and compensation. Unlike conventional TEQ, the equalizer is divided into two-stage structures. The first-stage equalizer is an adaptive multi-resampling equalizer, which carries out channel pre-equalization and Doppler rough estimation. This equalizer detects the equalization coefficient and selects the most matching multi-sampling rate interval for the output, to achieve an adaptive resampling. The second-stage equalizer is a parallel turbo equalizer, which sends the multi-sampling signal output by the first-stage equalizer to the turbo equalizer for joint equalization and decoding. Verified by simulations and lake experiments, the equalizer can effectively carry out Doppler estimation and compensation, ensure robust and reliable UAC in a Doppler environment, and does not require auxiliary signals for timescale estimation, greatly saving time and energy costs. The contribution of this article is as follows:

- (1) A two-stage adaptive multi-sampling turbo equalizer is proposed, which realizes robust and reliable underwater acoustic communication by combining an adaptive multi-resampling equalizer and parallel turbo equalizer.
- (2) An adaptive resampling method is proposed to adaptively update the resampling rate of the equalizer by detecting the equalization coefficient, to achieve Doppler estimation and compensation.

The rest of this article is organized as follows. Section 2 introduces the model of a single-carrier (SC) UAC system and describes the structure of the proposed equalizer. Section 3 presents and analyzes the proposed adaptive resampling method. Simulation and experimental results are presented in Section 4 to evaluate the performance of the proposed equalizer compared with the conventional TEQ based on timescale estimation. Finally, a conclusion is drawn in Section 5.

2. System Model

2.1. Signal Model

In an underwater channel, the signal received by the communication receiver is the sum of multiple proportional copies of the transmitted signal, each with a different delay and attenuation. Assuming that the difference in the angle of incidence reached by the multiple sound lines generated by the multipath effect is small, each path has the same Doppler factor μ_p , N_p is the number of paths, and A_p and τ_p are the attenuation coefficient and delay of the p th path; then, the time-domain shock response of the time-varying coherent multipath channel can be expressed as

$$h(\tau_p, t) = \sum_{p=1}^{N_p} A_p(t) \delta[t - \tau_p(t)] = \sum_{p=1}^{N_p} A_p(t) \delta[(1 + \mu_p)t - \tau_p] \quad (1)$$

where the Doppler factor $\mu_p \approx v_p/v_c$, and v_p and v_c are the relative velocity and underwater acoustic propagation velocity, respectively. The Doppler factor is on the order of 10^{-3} or 10^{-4} [32].

This paper uses binary phase shift keying (BPSK) modulation, and the SC signal transmitted in the passband can be expressed as

$$s(t) = \text{Re} \left\{ \sum_{k=0}^{K-1} x_k g(t - kT_0) e^{j2\pi f_c t} \right\}, t \in [0, KT_0] \quad (2)$$

where $\text{Re}\{\cdot\}$ is the real part, x_k is the k th symbol modulated by the BPSK (the total number of symbols is K), $g(t)$ is the raised cosine pulse waveform, f_c is the carrier frequency, and T_0 is the duration of a symbol. After undergoing an underwater acoustic channel, the SC signal is affected by the time-varying coherent multipath channel and ambient noise. Therefore, the received signal $\tilde{r}(t)$ can be expressed as

$$\tilde{r}(t) = h(\tau_p, t) * s(t) + \tilde{w}(t) = \sum_{p=1}^{N_p} A_p(t) s[(1 + \mu_p)t - \tau_p] + \tilde{w}(t) \quad (3)$$

where $\tilde{w}(t)$ is the additive ambient background noise. The received signal is down-regulated, and a low-pass filter (LPF) is used to obtain the baseband SC received signal, which is expressed as

$$\begin{aligned} r(t) &= \text{LPF} [\tilde{r}(t) e^{-j2\pi f_c t}] \\ &\approx \sum_{p=1}^{N_p} \left\{ A_p(t) e^{j2\pi f_c (\mu_p t - \tau_p)} \sum_{k=0}^{K-1} x_k g[(1 + \mu_p)t - kT_0 - \tau_p] \right\} + w(t) \end{aligned} \quad (4)$$

Therefore, the signal in the SC communication system is modeled as a multipath time-varying underwater acoustic channel affected by the Doppler Effect and environmental noise. In this paper, the transmission signal consists of a BPSK modulation signal.

2.2. Direct Adaptive Multi-Resampling Turbo Equalizer

Based on the above transmission signal model, this paper proposes the DAM-TEQ, which adaptively realizes the processing of the transmitted signal through a two-stage structure. The two-stage structure of the DAM-TEQ is shown in Figure 2. The first stage is an adaptive multi-resampling equalizer, and the second stage is a parallel turbo equalizer.

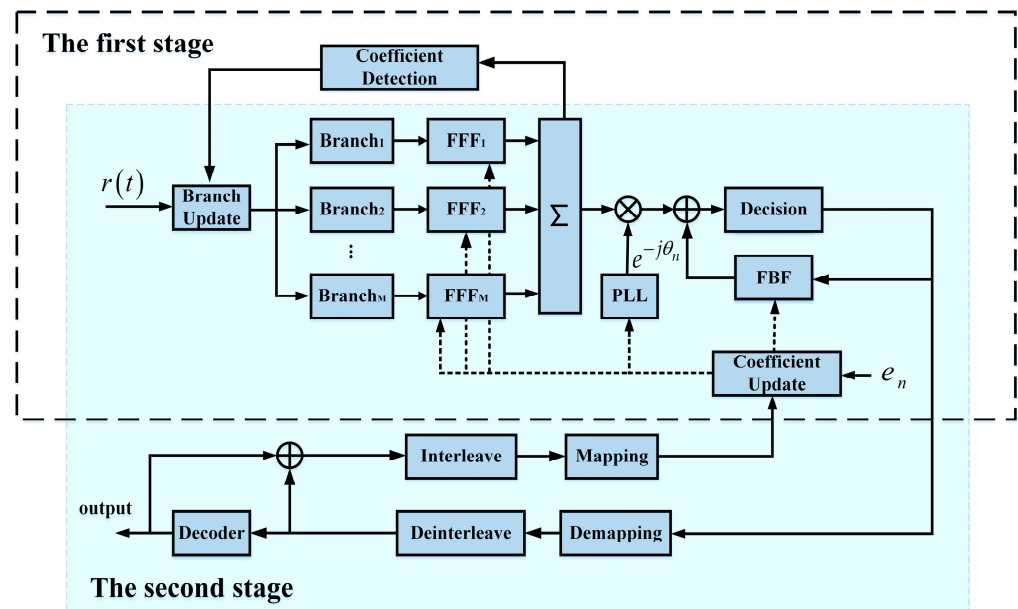


Figure 2. The structure of DAM-TEQ.

2.2.1. Adaptive Multi-Resampling Equalizer

Firstly, the received signal $r(t)$ is resampled by using the resampling factor $u_1, u_2, \dots, u_M$, and M resampled baseband signal branches are generated. Each resampling factor u_m

correspondingly represents an estimated Doppler factor μ_m . The m th resampling branch is denoted as

$$r_m(t) = r\left(\frac{t}{1 + u_m}\right) \quad (5)$$

Let the initial sampling rate be f_s , and the branch signal received at time n can be discretely expressed as

$$r_{m,n} = r_m[n] \approx \sum_{l=0}^{L-1} h_{m,l,n} x_{n-l} + w_{m,n} \quad (6)$$

$$h_{m,l,n} = h_l\left(\frac{n}{(1 + u_m)f_s}\right), w_{m,n} = w\left(\frac{n}{(1 + u_m)f_s}\right) \quad (7)$$

where x_{n-l} is the transmitted signal at time $(n - l)$, $h_{m,l,n}$ is the l th coefficient of the channel impulse response, and L is the total length of the channel impulse response. The resampled branches are fed into the DFE bit by bit, which consists of the m parallel feedforward filter (FFF), phase lock loop (PLL), and feedback filter (FBF). The equalizer coefficients include the FFF coefficient c_m , the phase correction coefficient θ_n , and the FBF coefficient b . The branch signals are processed sequentially, module by module.

Suppose the length of the FFF is $(N_1 + N_2 + 1)$, where N_1 , N_2 are the length of the non-causal and causal parts of the filter. Therefore, the m th FFF coefficient is expressed as

$$c'_m = [c_{-N_1}^m, \dots, c_0^m, \dots, c_{N_2}^m] \quad (8)$$

The input $r_{m,n}$ and output $p_{m,n}$ of the m th FFF are represented as

$$r'_{m,n} = [r_{m,n-N_1}, \dots, r_{m,n}, \dots, r_{m,n+N_2}] \quad (9)$$

$$p_{m,n} = c_{m,n}^H r'_{m,n} \quad (10)$$

PLL corrects the phase to obtain the correction coefficient θ_n .

Assuming the FBF length is N_3 , the FBF coefficient is expressed as

$$b' = [b_1, \dots, b_{N_3}] \quad (11)$$

The input $\widetilde{x}_n$ and output q_n of the FBF are represented as

$$\widetilde{x}'_n = [\widetilde{x}_{n-1}, \dots, \widetilde{x}_{n-N_3}] \quad (12)$$

$$q_n = b'^H \widetilde{x}'_n \quad (13)$$

Therefore, the output of this equalizer is

$$\hat{x}_n = \sum_{m=1}^M p_{m,n} e^{-j\theta_n} - q_n \quad (14)$$

The estimation error $e_n = x_n - \hat{x}_n$, and then recursive least squares (RLS) is used to solve, and the equalizer coefficients c_m , θ_n , and b are iteratively updated bit by bit according to the estimation error [33]. The resampling rate and the branches are updated by detecting the convergent equalizer coefficient, described in the next section.

2.2.2. Parallel Turbo Equalizer

The second-stage equalizer is the turbo equalizer, which is different from the first-stage equalizer in that it uses soft symbols to achieve decision feedback equalization, and the equalized symbols are passed between the equalizer and the decoder in the form of extrinsic information [34]. Due to the lack of accuracy of the Doppler factor coarsely

estimated by the first-stage equalizer, the second-stage equalizer also adopts the method of parallel multi-resampling, and the M-resampled branches selected by the last structure are combined as inputs, and these branches near the correct Doppler factor provide more frequency shift bases for the equalizer [35]. The result of the equalizer output is

$$\hat{x}_n = \sum_{m=1}^M c_{m,n}^H r_{m,n} e^{-j\theta_m} - \mathbf{b}^H \bar{\mathbf{x}}_n \quad (15)$$

where $\bar{\mathbf{x}}_n = [\bar{x}_{n-1}, \dots, \bar{x}_{n-N_3}]'$, and $\bar{x}_n$ is the prior mean at time n , which is expressed as

$$\bar{x}_n = \sum_{\alpha_{n,i} \in S} \alpha_{n,i} P(x_n = \alpha_{n,i}) \quad (16)$$

where S is the corresponding set of symbols of the symbol x_n and $\alpha_{n,i} = 1$ or -1 for BPSK modulation. In BPSK modulation, each bit is mapped to a symbol, where symbols 1 and -1 correspond to bits 1 and 0. In detail, 1, 0 to 1, -1 is achieved through mapping, and 1, -1 to 1, 0 is achieved through demapping. In addition to the judgment result, the equalizer also outputs the extrinsic information $L_e^E(\alpha_{n,i})$, which is used for transmission and iteration with the decoder, and it is expressed as

$$\begin{aligned} L_e^E(\alpha_{n,i}) &= \underbrace{\ln \frac{P(\alpha_{n,i} = 1 | \hat{x}_n)}{P(\alpha_{n,i} = -1 | \hat{x}_n)}}_{\text{Posterior LLR}} - \underbrace{\ln \frac{P(\alpha_{n,i} = 1)}{P(\alpha_{n,i} = -1)}}_{\text{Prior LLR}} \\ &= \ln \frac{p(\hat{x}_n | \alpha_{n,i} = 1)}{p(\hat{x}_n | \alpha_{n,i} = -1)} \end{aligned} \quad (17)$$

$$p(\hat{x}_n | \alpha_{n,i}) = \frac{1}{\pi \delta_n^2} \exp\left(-\frac{|\hat{x}_n - \mu_n \alpha_{n,i}|^2}{\delta_n^2}\right), \alpha_{n,i} = 1 \text{ or } -1 \quad (18)$$

In general, assuming that the channel impulse response remains unchanged within the length of a frame of data, the length is N_s , and the channel parameters μ'_n and δ_n^2 are approximated by time averaging:

$$\mu'_n = \frac{1}{N_s} \sum_{n=1}^{N_s} \frac{\hat{x}_n}{Q(\hat{x}_n)} \quad (19)$$

$$\delta_n^2 = \frac{1}{N_s} \sum_{n=1}^{N_s} |\hat{x}_n - \mu'_n Q(\hat{x}_n)|^2 \quad (20)$$

where $Q(\hat{x}_n)$ is the decided value of $\hat{x}_n$. The extrinsic information $L_e^E(\alpha_{n,i})$ output by the equalizer is obtained by (15)–(20), and the prior information $L_a^E(\alpha_{n,i})$ input to the decoder is obtained after demapping and deinterleaving. The decoder described in [36] is used for decoding, and the process is not described in detail. The decoder obtains two outputs: the decision information bit result $\hat{d}_n$ and the decoder extrinsic information $L_e^D(\alpha_{n,i})$. The extrinsic information passes through the S16 interleaver and mapping to obtain the prior information $L_a^D(\alpha_{n,i})$ input to the equalizer, which is used to calculate the prior probability $P(x_n = \alpha_{n,i})$ in (16):

$$P(x_n = \alpha_{n,i}) = \frac{1}{2} \left[1 + \alpha_{n,i} \tanh \frac{L_a^D(\alpha_{n,i})}{2} \right] \quad (21)$$

With the iteration, the accuracy of the prior information gradually improves, and finally the joint decoding effect of the turbo equalizer is achieved. The Doppler estimation, compensation, and communication reception and decoding process are realized through the cascading of first- and second-stage equalizers.

3. Adaptive Resampling Method

3.1. Adaptive Process

The proposed DAM-TEQ realizes the adaptive process by using the resampling rate update method based on equalization coefficient detection, as shown in the coefficient detection and branch update blocks in Figure 2. The adaptive process consists of initialization, an adaptive update of the center resampling rate and branch interval, and the selection of an appropriate resampling rate interval, and the implementation process is shown in Figure 3.

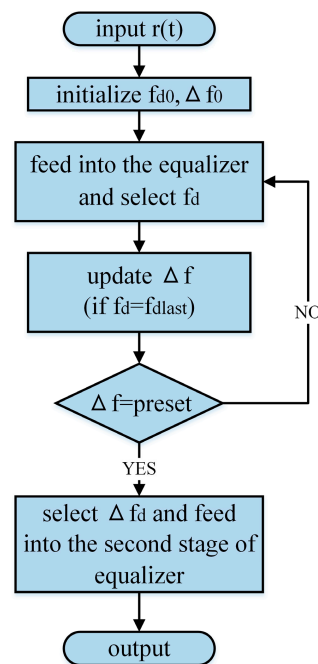


Figure 3. Adaptive process.

According to (10) and (14), it can be seen that in the first stage of the DAM-TEQ, the equalizer output is composed of the M parallel FFF and FBF outputs, where M FFF provides received signal branches with different Doppler shifts. In the process of parallel equalization, the content of each branch in the final output $\hat{x}_n$ is determined by the equalization coefficient after convergence. The Doppler Effect may worsen when the resampling factor u_m deviates from the correct Doppler factor, so this branch should occupy a small proportion of $\hat{x}_n$ to ensure a small estimation error and a good equilibrium effect. Based on this fact, it is possible to determine the detection method of the first-stage of the equalizer: the most suitable resampling branch is selected by judging the amplitude of the FFF coefficient, which is expressed as

$$[|c_{max}|, m_{max}] = \max(|c_1|, \dots, |c_M|) \quad (22)$$

where $|c_{max}|$ is the FFF coefficient with the largest amplitude, m_{max} is the branch's subscript, and the branch's resampling rate is f_d . Let f_d be the central resampling rate, and then select the estimated Doppler factor μ_{max} by updating f_d and narrowing the sampling rate interval Δf , continuously changing the resampling factor and retaining the resampling branch with a large coefficient amplitude. To facilitate the study, the number of resampled branches $M = 3$. The specific implementation steps are as follows:

Step 1: Initialize the resampling branch. Each resampling factor μ_m corresponds to a resampling rate f_m , and firstly, the initial central sampling rate f_{d0} (generally using the initial sampling rate f_s) and the initial resampling rate interval Δf_0 are set, then the resampling rates of the three branches are $f_{d0} - \Delta f_{d0}$, f_{d0} , $f_{d0} + \Delta f_{d0}$;

Step 2: Search and update the central resampling rate. The first-stage equalizer processes the parallel input to detect the equalization coefficient, and selects the one with the largest amplitude of the FFF coefficient in the three branches as the central branch, and its sampling rate is updated to the central sampling rate f_d ;

Step 3: Update the adaptive resampling rate interval Δf . If the center frequency is the same as the previous search, the Δf is reduced. If it is different, the Δf is kept the same and the resampling rates for the three updated branches are $f_d - \Delta f, f_d, f_d + \Delta f$. Repeat steps 2 and 3 until the Δf is narrowed down to a preset minimum to achieve a coarse Doppler estimate;

Step 4: Select the resampling rate interval Δf_d . In order to provide a frequency-shift basis for the parallel input of the second-stage equalizer, re-select the appropriate sampling rate interval Δf_d and send the three branches of $f_d - \Delta f_d, f_d, f_d + \Delta f_d$ to the multi-resampling turbo equalizer for joint equalization decoding.

3.2. Complexity Analysis

Compared with the conventional TEQ, the DAM-TEQ using the adaptive resampling method increases the complexity in two aspects: the adaptive process of the first-stage structure and the multiple parallel inputs of the second-stage structure. The rest of the structure is consistent with the conventional TEQ. Therefore, for ease of calculation and generality, the analysis is mainly based on the equilibrium complexity of one frame of data (concentrated on the RLS), which does not include the decoding process. The number of computations required for a single information symbol is used to characterize the complexity of the algorithm. In the calculation, the FFF length $N_a = N_1 + N_2 + 1$ and the FBF length $N_b = N_3$.

In the conventional TEQ, $3(N_a + N_b)^2 + 3(N_a + N_b)$ times multiplication and $3(N_a + N_b)^2 + N_a + N_b$ times addition need to be calculated, and the overall complexity is $O[(N_a + N_b)^2]$. It is assumed that the resampling rate is updated N times in the adaptive process, and there are M branches of parallel input in the DAM-TEQ. Compared with the conventional TEQ, the proposed equalizer needs to calculate the number of equalizations additionally as follows: N adaptive processes in the first-stage structure, including the calculation of M FFF branches each time, and $M-1$ FFF branches in the second-stage structure. The added complexity of the proposed equalizer is shown in Table 1. The overall complexity of the equalizer is $O[(MN_a + N_b)^2]$, and the sacrifice of this part of the complexity relative to the performance improvement can be accepted. At the same time, due to the relatively low number of symbols transmitted in long-range UAC, the overall complexity will not be a limiting factor for the equalizer.

Table 1. Increased complexity.

Structure	Increased Multiplications	Increased Additions
DAM-TEQ _(I)	$3N(MN_a + N_b)^2 + 3N(MN_a + N_b)$	$3N(MN_a + N_b)^2 + N(MN_a + N_b)$
DAM-TEQ _(II)	$3(M-1)^2N_a^2 + 3(M-1)N_a$	$3(M-1)^2N_a^2 + (M-1)N_a$

4. Performance Evaluations

4.1. Simulations

In order to verify the performance of the proposed DAM-TEQ, a simulation analysis is carried out in this paper. The simulation will be conducted in three aspects: feasibility, reliability, and comparative performance.

4.1.1. Feasibility

The underwater acoustic simulation channel parameters are set as follows: paths number $N_p = 15$, the average time interval $\Delta\tau$ between the two paths is 2 ms, the power difference ΔP_p of the closing path is 20 dB, and the average sound velocity $v_c = 1500$ m/s.

The relative speed of platform movement between the transmitter and receiver is v_p and the standard deviation σ_v of this velocity is 0.01 m/s, evenly distributed in each path. Table 2 shows the parameters of the transmitted SC signal; 1296 data bits are used as the communication sequence, which are modulated into a carrier with a frequency of 450 Hz after convolutional coding. In BPSK modulation, the bits and symbols are one-to-one, which means that the bit rate is equal to the baud rate, and this number is set to 200 in the simulation. Therefore, the channel bandwidth is 200 Hz. The SC signal sampling rate is set to 40,000 Hz, which can be applied to the transducer.

Table 2. SC signal parameters.

Parameter	Value
Carrier frequency f_c	450 Hz
Modulation mode	BPSK
Sequence length	1296 bits
Baud rate	200 Baud
Bandwidth	200 Hz
Initial sampling rate f_s	40,000 Hz
Convolutional coding	(2, 1, 3)

Taking the signal-to-noise ratio (SNR) = 0 dB and $v_p = 1$ m/s as an example, the process of using the DAM-TEQ to process the simulation data is analyzed. In the first stage of the equalizer, set $f_{d0} = 40,000$, $\Delta f = 50$, and $\Delta f_{\min} = 5$. The updated results of the equalization coefficient detection and resampling branches in the adaptive process are shown in Figure 4. The resampled branch with a relatively large amplitude of the equalization coefficient was selected as the central branch, and the adaptive sampling rate interval was narrowed if the central branch remained unchanged. The coarse Doppler estimation is realized after a few iterations, and the final value of f_d is 39,975. Select the resampling rate interval Δf_d , and generate three branches, where the resampling rates are $f_d - \Delta f_d$, f_d , and $f_d + \Delta f_d$. Then the resampled signals from the three branches are fed into the second-stage equalizer. The processing result is 0 BER, and the feasibility of the proposed equalizer is verified by simulation.

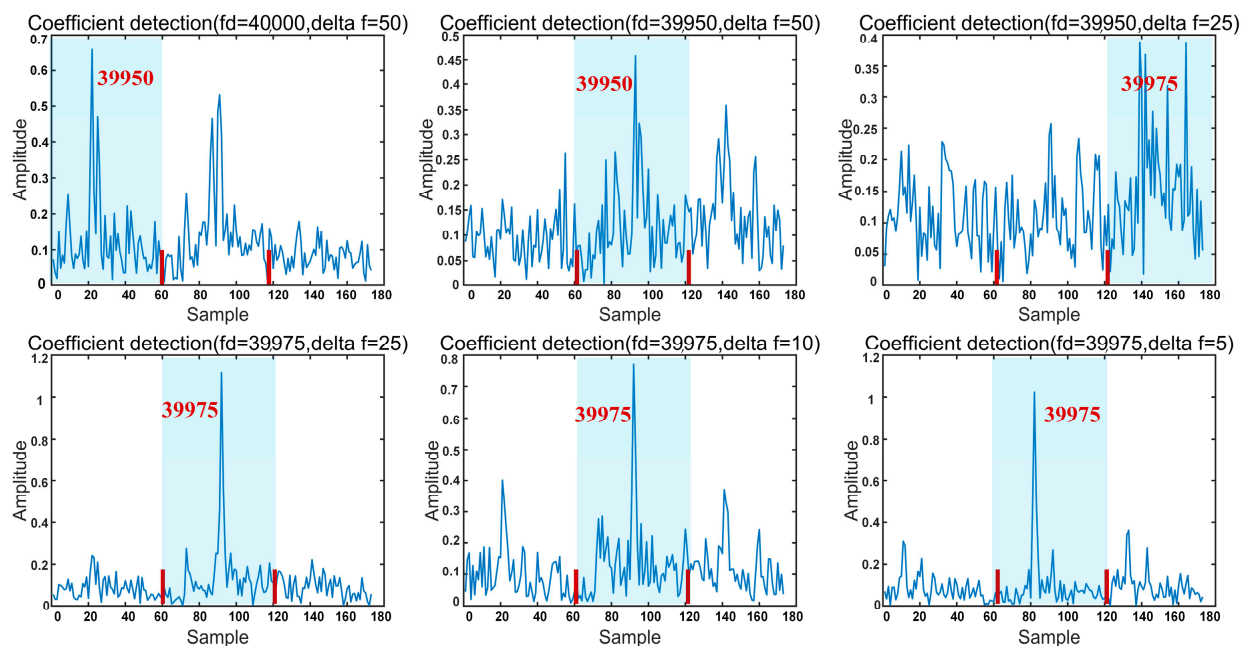


Figure 4. Coefficients and branch update processes.

In order to further verify the effectiveness of the DAM-TEQ, the simulation is carried out under three different frequency-shift conditions, $v_p = 1$ m/s, $v_p = 2$ m/s, and $v_p = 3$ m/s, and the frequency shifts are calculated by $\Delta f_{Doppler} = f_c v_p / v_c$. The simulation results are shown in Figure 5. The simulation results show that the BER with the small relative moving speed is relatively low. Still, the BER decreases with the increase in the SNR under different frequency-shift conditions, indicating that the proposed equalizer has an effective Doppler estimation and compensation ability under different frequency-shift conditions.

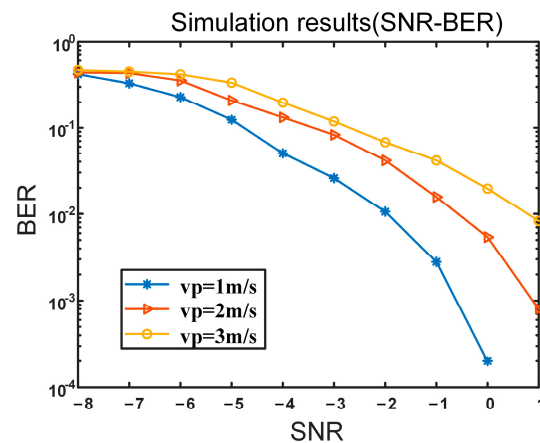


Figure 5. DAM-TEQ processing results under different relative speeds.

4.1.2. Reliability

Under the above conditions, the processing results of different SNRs are simulated to verify the reliability of the proposed DAM-TEQ. In order to obtain convincing statistical rules, 100 simulations were carried out for different SNR groups, which effectively avoided the contingency of the results and increased the reliability and generality, as shown in Figure 6. The criterion for determining whether the information transmission is accurate is indicated by a processed BER of the order of magnitude α (a BER at or below α is considered to be an accurate transmission). The figure shows the accuracy of different criteria at different SNRs. When $\alpha = 10^{-2}$, the accuracy can reach more than 90% after the SNR reaches -2 . When $\alpha = 10^{-3}$, the accuracy can be maintained at more than 90% after the SNR reaches 3. When $\alpha = 10^{-4}$, the accuracy can be maintained above 90% after the SNR reaches 6. The results show that the proposed DAM-TEQ can demonstrate a high transmission accuracy at a low SNR, and with the increase in the SNR, the accuracy will continue to remain at a high level, which can ensure stable and reliable information transmission.

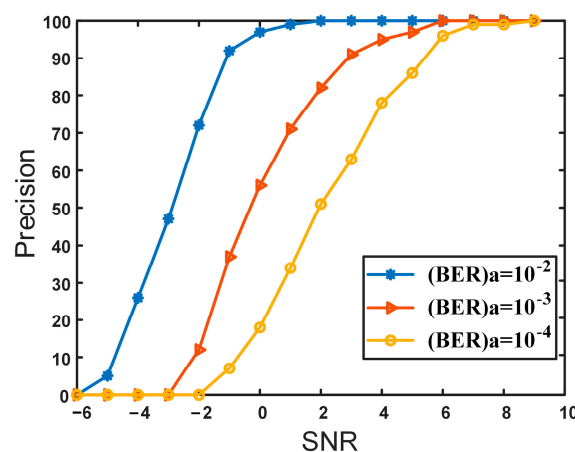


Figure 6. Accuracy statistical results.

4.1.3. Comparison

In order to compare and analyze the performance of the DAM-TEQ and the conventional TEQ based on timescale estimation under different underwater acoustic channel conditions, a comparative simulation is set up on the basis of the above simulations: $v_p = 1$ m/s, S1 is $N_p = 10$ and $\Delta\tau = 2$ ms, which simulates the situation that the channel is stable and good, and the relevant peaks of the LFM signal can be accurately obtained; and S2 is $N_p = 20$ and $\Delta\tau = 5$ ms, in which there is a propagation path similar to the energy level. It simulates the situation that the channel is bad, the multipath and attenuation are serious, and the relevant peaks cannot be accurately obtained. Under different SNRs, two equalizers are used to process the communication signals in different situations, and the processing results are displayed through the BER performance curve, as shown in Figure 7.

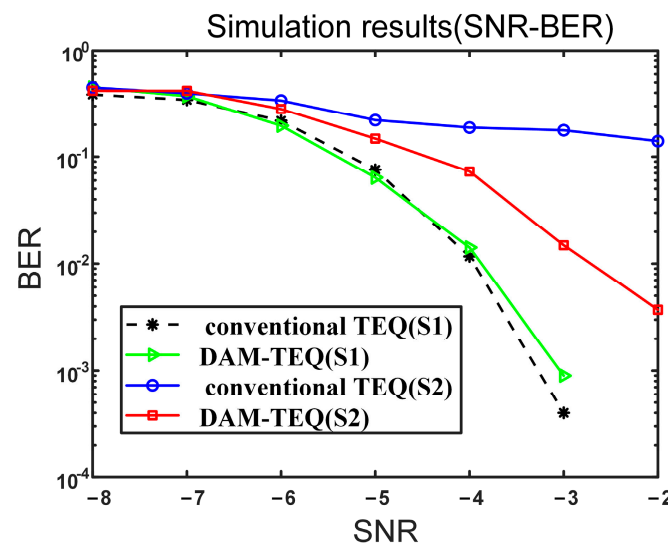


Figure 7. Comparison of equalizer performance under different conditions.

The simulation results show that under the S1 condition, the conventional TEQ and the DAM-TEQ show excellent performance due to the accurate LFM signal correlation peak, and the conventional method works on an accurate timescale. However, under the condition of S2, the conventional TEQ cannot work accurately due to the influence of a poor channel, and the BER is maintained at a high level. In contrast, the proposed equalizer can still work normally, and with the increase in the SNR, the BER gradually decreases. The BER performance is significantly better than that of the conventional TEQ. The comparative analysis results verify the effectiveness and good performance of the DAM-TEQ.

4.2. Lake Experiment

In order to further verify the performance of the proposed turbo equalizer, a lake experiment was carried out in Dongjiang Lake on 6 July 2023, and the experimental scene layout is shown in Figure 8. The underwater communication distance is 6 km, the transmitting transducer and the receiving hydrophone are at a water depth of 30 m, the two platforms move relative to each other, and the parameters of the SC signal transmitted in the experiment are consistent with the simulation.

To compare and analyze the proposed turbo equalizer, the communication experimental data were processed accordingly, and three experimental scenarios were simulated in which the correlation peaks could be accurately identified, the correlation peaks could not be identified, and with the presence of multiple correlation peaks. The corresponding times of these three scenarios correspond to 11:54:45, 11:56:25, and 11:58:05, respectively. Three sets of typical experiments are selected for detailed comparison and analysis, and the cross-correlation results between the communication waveform and the LFM signal are

shown in Figure 9. The experiments are processed by two methods: the conventional TEQ based on timescale estimation and the proposed DAM-TEQ.

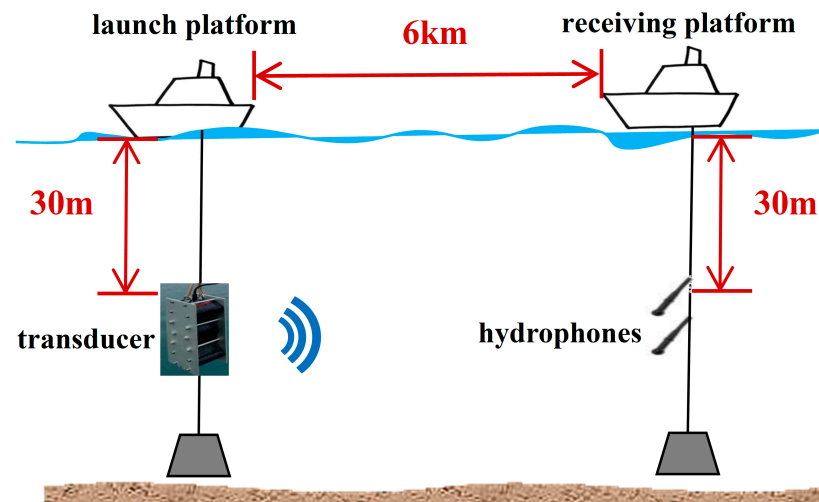


Figure 8. Experimental setup.

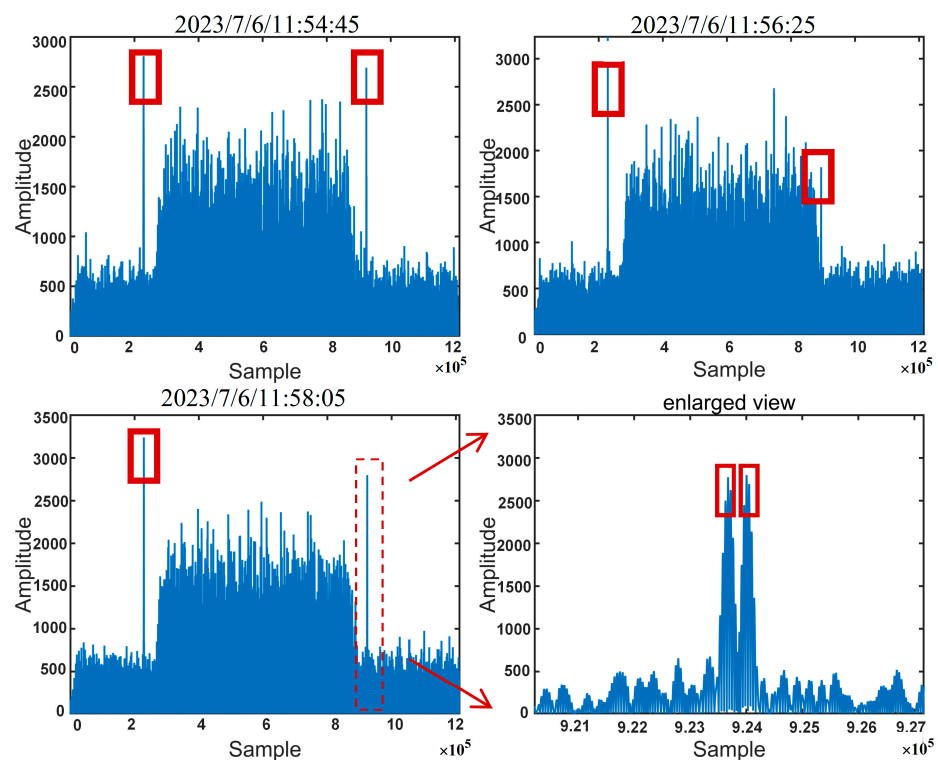


Figure 9. Correlation output between the communication signal and the LFM.

For the three sets of experiments, the constellation diagram obtained by the two equalizer processing methods is shown in Figure 10. In the first set of experiments, the precise position of the synchronization signal before and after the signal frame can be obtained by the method of cross-correlation and maximum search, and the received signal can be accurately processed by using the traditional Doppler estimation compensation method and turbo equalization, and the results show no BER. At the same time, the constellation diagram conforms to the characteristics of the BPSK-modulated signal, as shown in (a), (b) in Figure 10. However, in the second and third sets of experiments, due to the influence of complex time-varying channels such as multipath effects, the

phenomenon of weak correlation peaks or multiple correlation peaks appeared, and the accurate correlation peak position could not be obtained, resulting in the failure of the conventional method to work on an accurate timescale. Doppler compensation cannot be used to estimate the signal frame duration accurately. The processing results show a high BER, and the constellation diagram distribution is chaotic, as shown in (c), (d), (e), and (f) in Figure 10. In this method, the decoding performance cannot be improved after multiple iterations of turbo equalization, which means that turbo equalization fails. Correspondingly, the DAM-TEQ can adaptively update the resampling branch through the coefficient detection method, realize a Doppler estimation and compensation, and obtain a better BER performance for the three sets of experiments. It can be seen from the constellation diagram of the three sets of experiments that there is a slight error in the Doppler rough estimation of the first stage of the proposed equalizer, which causes the constellation diagram to rotate slightly, but it is still within an acceptable range, which can be corrected by parallel turbo equalization. At the same time, the compensated signal can be decoded by multiple iterations of the turbo equalizer to achieve error-free information transmission. The results of each group of experiments were compared and presented by BER (no iteration), as shown in groups 1, 3, and 5 in Table 3. In addition, the table also records the number of branch updates and iterations with a BER of 0 after using the DAM-TEQ.

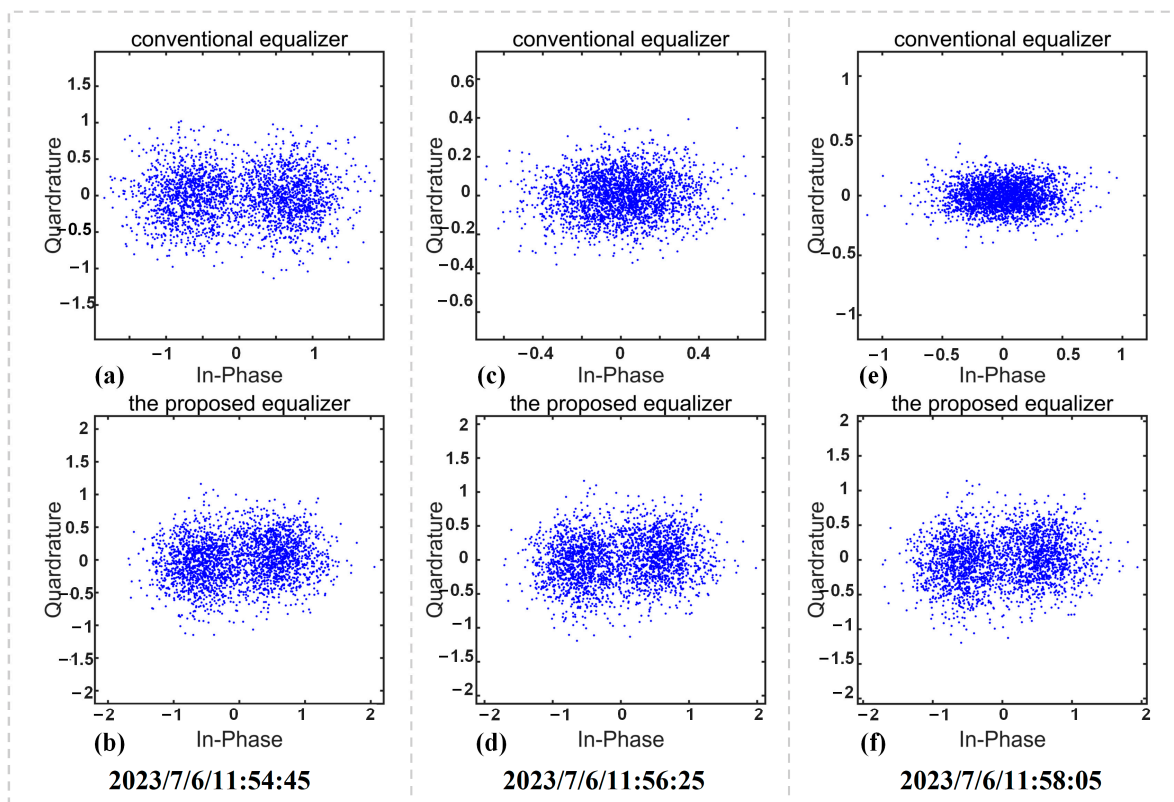


Figure 10. The constellation diagrams processed by the two equalizers.

Since the experimental data were collected at a high SNR, the performance of the DAM-TEQ was also verified by adding additional Dongjiang Lake environmental noise to the original data during processing, as shown in groups 2, 4, and 6 in Table 3. After the SNR is reduced by about 5 dB, the BER performance trend is basically unchanged, and only group 4 cannot eliminate the interference effect of noise by the equalizer due to the low SNR. The results show that the DAM-TEQ maintains a great advantage and that some experimental groups can maintain the information transmission without error by increasing the number of iterations. In addition, a set of communication signals that were

heavily disturbed was also processed, and the associated peaks could not be accurately estimated, as shown in group 7 in Table 3. Although the final BER is still relatively high, the performance of the proposed equalizer is greatly improved compared with the conventional method, which further illustrates the advantages of the DAM-TEQ.

Table 3. Experimental results.

Group	Recorded Data	BER of the Conventional TEQ	BER of the DAM-TEQ	Update Times	Iteration Times
1	11:54:45	0	0	4	1
2	11:54:45(−5 dB)	0	0	4	3
3	11:56:25	0.4861	0	4	4
4	11:56:25(−5 dB)	0.4961	0.1690	4	−
5	11:58:05	0.4846	0	4	2
6	11:58:05(−5 dB)	0.4468	0	4	4
7	11:53:05	0.4653	0.1705	4	−

5. Conclusions

In this paper, an adaptive multi-resampling turbo equalizer was proposed to solve the Doppler problem, and the equalizer's feasibility and bit error rate performance were verified by simulation and lake experiments. The simulation and experimental results showed that the proposed turbo equalizer can maintain an excellent performance under conditions where the traditional method cannot work normally due to poor channels. At the same time, the proposed processing method no longer requires the relevant signals for timescale estimation to be inserted before and after the signal frame, which not only greatly reduces the transmission time and power consumption but also effectively improves the reliability of UAC.

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